STORY WALK APP - TECHNICAL

Keep the first version deliberately modest: a single-route, offline-first audio storytelling app, rather than trying to build a fully featured walking-tour platform straight away.

The key technical challenge is making the story feel as though it unfolds around the walker, while ensuring it still works reliably in a rural area with patchy mobile reception. The design below gives Felix something practical to work towards, while also providing material for the 3-Minute Arts funding proposal.

Sarkless Kitty Story Walk

Technical specification – prototype application

Prepared for: 3-Minute Arts CIC
Proposed developer: Felix Chalton-Thivillon
Project: Farndale Story Walk, North York Moors
Development period: 2027–28, subject to funding

1. Project overview

The aim is to develop a smartphone application that delivers an immersive, interactive audio story as people follow a designated walking route in Farndale.

As walkers reach particular locations, the app will play sections of a story, using recorded voices, narration, music and sound effects. At selected points, users may be invited to make a choice or respond to a prompt, influencing what they hear next.

The application should be designed to work without a mobile internet connection once the story has been downloaded. This is particularly important given the limited and variable mobile coverage in parts of the North York Moors.

The initial prototype will focus on one route and one story. It should provide a foundation that could subsequently be adapted for other Story Walks across the North York Moors, coast and Cleveland.

2. Proposed prototype features

The first version should aim to include:

Location-based story playback

- A simple map or route display showing the walk and its story locations.
- GPS detection to identify when a walker approaches a designated point.
- Automatic playback of the relevant audio section, with a clear indication of the current story location.
- A manual option to start or replay a section if GPS detection is inaccurate.

Audio and interactive storytelling

- Playback of pre-recorded narration, character voices and sound effects.
- Audio sequences that can be triggered by reaching a location.
- Simple interactive choices or prompts at selected points, allowing the story to follow different paths.
- Pause, resume and replay controls, with volume adjustment.

Offline operation

- Download the complete story and associated audio before starting the walk.
- Store the content locally on the phone.
- Ensure that GPS triggering and audio playback continue to work without mobile data or Wi-Fi.

User experience

- A straightforward interface suitable for a wide range of ages and levels of technical confidence.
- Clear instructions for starting the walk and downloading the story.
- A simple way to recover if a walker misses a location or wants to hear a section again.

3. Suggested technical architecture

A relatively simple modular architecture should make the prototype easier to develop, test and extend.

A. Mobile application

Suggested technology: **Flutter (Dart)**.

Flutter is a cross-platform framework that can be used to develop applications for Android and iOS from a shared codebase. This would provide a possible route to supporting both platforms without developing two entirely separate applications.

For the initial prototype, Felix could concentrate on one platform if that makes development and testing more manageable.

B. Location service

Suggested technology: Flutter location/GPS packages, such as `geolocator`.

- Store the latitude and longitude of each story location.
- Periodically check the phone's GPS position.
- Calculate the distance between the walker and the next story point.

- Trigger the appropriate story section when the walker enters a defined radius.

The trigger radius should be configurable and tested on the actual route. GPS accuracy can vary considerably, particularly near trees, buildings and uneven terrain.

A manual playback option should always be available as a fallback.

C. Audio playback

Suggested technology: `just_audio` or a comparable Flutter audio package.

- Play locally stored audio files.
- Support narration, character dialogue, music and sound effects.
- Allow playback to be paused, resumed and replayed.
- Manage transitions between story sections.

For the first prototype, audio files could be included within the application itself. This would avoid the need to develop a separate download system immediately.

D. Story structure and branching

The story should be stored separately from the main application code, using a simple structured format such as JSON.

Each story section could contain:

- A unique identifier.
- The location or trigger associated with it.
- The audio file to be played.
- Any accompanying text or instructions.
- Optional choices and the identifiers of the next sections.

This would allow the story content to be changed or extended without rewriting the main application.

A simple branching structure would be sufficient initially; a sophisticated narrative engine is not necessary.

E. Offline content storage

For the prototype, audio and story data can be stored locally on the device.

A later version could include a download facility, allowing users to download different Story Walks before setting out. Packages of audio and story data could be downloaded together and stored locally.

The application should check that the content is available before the walk begins.

4. Simplified system flow

1. The user downloads or opens the Sarkless Kitty story.
2. The app loads the route and its story locations.
3. The user starts the walk.
4. The app checks the phone's GPS position.
5. When the user reaches a designated location, the corresponding audio section is triggered.
6. The user listens to the story and, where appropriate, responds to a prompt or makes a choice.
7. The app selects the next story section and continues as the walk progresses.

If GPS triggering fails, the user can select the relevant story section manually.

5. Development stages

Stage 1 – Requirements and story mapping

- Agree the walking route and identify the principal story locations.
- Work with the creative team to establish the sequence of story sections.
- Identify where audio should be triggered and where interactive choices might be used.
- Establish the minimum features needed for the first prototype.

Output: a route map, story flow diagram and basic technical specification.

Stage 2 – Basic application prototype

- Set up the development environment and Flutter project.
- Create a basic user interface.
- Add the route locations and GPS detection.
- Implement audio playback using sample recordings.
- Provide manual controls to select and replay story sections.

Output: a working demonstration of location-triggered audio playback.

Stage 3 – Interactive and offline functionality

- Introduce the story structure and simple branching choices.
- Add locally stored audio and story content.
- Test operation with mobile data and Wi-Fi switched off.
- Add basic error handling and recovery options.

Output: a working prototype of the interactive, offline audio experience.

Stage 4 – Field testing and refinement

- Test the application along the actual Farndale route.
- Check GPS accuracy and the reliability of location triggers.
- Adjust trigger distances and timing.
- Test audio playback, battery consumption and usability.
- Gather feedback from a small group of walkers and make improvements.

Output: a tested prototype suitable for demonstration and further development.

Stage 5 – Future development

Subject to the success of the prototype and further funding:

- Develop a more polished version for public use.
- Support additional Story Walks and different story packages.
- Introduce a straightforward content-management process.
- Improve accessibility and provide additional language or text options if required.
- Consider wider distribution through the relevant app stores.

6. Key technical considerations and risks

Issue	Proposed approach
Poor mobile reception	Download all story content in advance; operate offline
Inaccurate GPS	Allow manual selection and replay of story sections
Battery consumption	Avoid unnecessarily frequent GPS checks; test on the route
Different phone models	Test on a small range of Android/iOS devices
Walkers moving at different speeds	Avoid relying on precise timing; trigger by location
Missed story locations	Allow users to return to previous sections
Changing story content	Keep story data separate from the application code

7. Initial prototype scope

The first prototype should demonstrate the central idea rather than attempt to deliver a finished commercial application.

A successful demonstration would show:

- A short section of the Farndale route.
- At least three GPS-triggered story locations.
- Recorded audio and sound effects.
- At least one interactive choice.
- Offline playback.
- Manual controls as a fallback.

This should be sufficient to demonstrate the feasibility of the concept, identify practical development issues and provide a basis for planning the full application.

8. Potential for wider use

The longer-term aim is to develop a reusable platform for community-led Story Walks. The underlying application could remain largely the same, with different routes, locations, recordings and story structures supplied as separate content packages.

This would make it possible for local groups to develop their own walks based on local folklore, history, landscape and personal stories, without having to commission an entirely new application each time.

A few recommendations before Felix starts

I would keep the technical ambition at the level described above. There are some tempting additions, but they could make the first version unnecessarily complicated.

1. Start with three locations, not the whole walk. A short demonstration with three GPS triggers, audio and one choice would prove the central idea. It would also give the funding application something concrete to describe.

2. Use sample recordings initially. Felix doesn't need to wait for actors, final scripts or sound effects. He can use temporary recordings to demonstrate how the system works.

3. Make offline operation a core requirement from the beginning. It is much easier to design for offline use at the outset than to add it later.

4. Don't make GPS the only way to navigate the story. A manual 'play this section' option is important. QR codes at story points might also be worth considering, although they would need to be incorporated into the physical trail and signage.

5. Keep the story content separate from the software. This is particularly important if 3MA eventually wants to develop several Story Walks. The aim would be for new stories to be added without Felix having to rebuild the application each time.

What Felix could put on his CV

At the moment, the project is at the technical scoping and proposal stage, so I would describe his contribution accurately:

Interactive Story Walk Application – 3-Minute Arts CIC
2026–present | Technical scoping and proposed development, subject to funding

Contributing to the technical design of an interactive, location-based audio storytelling application for a community heritage project in the North York Moors. Exploring an offline-first mobile application incorporating GPS-triggered audio, interactive story sequences and locally stored content. Developing a prototype specification covering application architecture, location services, audio playback and field testing.

Once he has built the prototype, he can replace this with a much stronger entry describing what he actually implemented, the technologies used and the results of testing.

Further thought: It would be idea if Felix could **make a small working demonstration before the funding decision**, if time allows. Even a **very simple app with three simulated locations and sample audio would make the proposal more tangible**, help identify technical problems early, and give him something demonstrable for his Cambridge Consultants application.